

ATSUGI AB, Japan

WMO: 476790

Lat: 35.455N

Lon: 139.450E

Elev: 215

StdP: 14.58

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 43319

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	30.3	32.2	6.6	7.8	40.5	10.0	9.3	39.9	25.2	43.8	22.8	44.3	5.2	0	0.471

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.6	91.7	78.1	89.7	77.1	87.8	76.7	79.4	87.6	78.6	86.3	77.8	85.3	11.6	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.2	142.8	83.2	76.6	139.9	83.1	75.3	133.8	82.9	43.1	87.9	42.2	86.8	41.3	85.0	84.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.6	20.8	18.5	DB	26.0	96.5	2.6	1.9	24.1	97.9	22.6	99.0	21.1	100.1	19.2	101.5	(4)
(5)			WB	21.7	81.5	1.9	1.3	20.3	82.4	19.1	83.1	18.1	83.8	16.6	84.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	61.5	42.2	43.5	49.1	58.4	66.6	72.2	79.5	81.8	75.5	66.0	55.7	46.6	
	DBStd	14.41	3.87	4.80	5.58	5.91	4.63	4.46	4.88	4.17	5.41	5.07	5.09	4.74	
	HDD50	667	246	191	87	7	0	0	0	0	0	0	9	127	
	HDD65	2953	708	602	493	208	37	4	0	0	2	48	281	570	
	CDD50	4866	2	9	59	259	514	665	913	985	764	495	179	22	
	CDD65	1678	0	0	0	10	86	219	448	520	316	78	1	0	
	CDH74	13465	0	0	0	33	301	1000	4097	5461	2296	276	1	0	
	CDH80	4326	0	0	0	2	31	192	1419	2031	615	36	0	0	
Wind	WSAvg	8.7	8.1	8.8	9.9	10.0	9.3	8.2	8.6	8.5	8.7	8.7	8.0	7.7	
Precipitation	PrecAvg	66.5	2.6	2.8	4.8	5.3	6.1	7.5	7.2	6.4	9.2	7.6	4.3	2.6	
	PrecMax	89.9	5.7	6.0	9.1	10.3	12.0	11.7	16.1	17.0	17.6	28.3	10.4	6.4	
	PrecMin	44.8	0.0	0.2	1.2	2.0	2.4	3.4	0.9	0.6	1.7	0.5	0.1	0.3	
	PrecStd	11.1	1.6	1.6	2.0	2.2	2.4	2.2	4.1	3.9	4.5	5.6	2.6	1.6	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.8	66.1	69.6	77.3	83.8	88.2	94.8	95.1	91.5	84.2	73.2	66.3	
		MCWB	51.7	55.2	56.3	61.6	66.3	74.3	78.2	79.3	76.1	69.5	62.1	56.4	
	2%	DB	55.8	60.7	65.9	73.6	80.2	84.5	91.5	92.8	88.1	79.5	69.9	61.1	
		MCWB	46.0	52.1	53.8	60.1	65.3	71.8	77.9	78.7	75.6	68.1	59.9	50.9	
	5%	DB	53.5	55.7	62.6	71.2	77.2	82.2	89.5	90.0	86.0	77.0	67.6	57.6	
		MCWB	43.4	45.9	52.8	59.2	64.2	70.8	77.3	77.6	74.4	66.8	57.7	48.1	
	10%	DB	50.4	52.2	59.4	68.2	75.1	79.2	87.4	88.2	83.8	74.8	64.6	55.4	
		MCWB	40.6	42.8	50.3	57.7	63.5	69.4	76.7	77.2	73.5	64.9	55.7	46.1	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.7	58.0	59.7	67.4	71.5	76.4	80.1	81.0	79.1	74.0	65.4	59.7
			MCDB	59.0	63.9	64.1	71.4	76.8	84.9	89.9	90.2	85.4	79.1	69.8	64.2
2%		WB	48.1	52.1	57.1	63.5	68.8	74.5	79.1	79.9	77.9	71.1	62.2	53.1	
		MCDB	54.4	59.3	63.0	70.6	75.4	81.8	88.0	88.1	84.0	77.2	67.3	59.6	
5%		WB	44.3	47.5	54.6	61.0	66.9	72.8	78.2	79.0	76.7	68.6	59.7	49.9	
		MCDB	51.4	54.0	61.3	68.0	74.0	79.3	86.5	86.7	82.9	75.0	64.7	56.0	
10%		WB	41.9	44.3	51.3	59.3	65.2	71.2	77.3	78.3	75.5	66.2	57.3	47.4	
		MCDB	49.1	51.2	58.4	66.1	72.4	77.4	85.1	85.7	81.7	73.1	63.0	53.9	
Mean Daily Temperature Range		5% DB	MDBR	15.6	15.2	15.1	14.9	13.7	11.4	11.3	11.6	11.8	12.2	13.7	15.1
			MCDBR	19.5	19.9	19.0	18.0	17.3	15.1	14.2	13.6	14.1	15.2	17.0	18.4
	MCWBR		13.2	14.2	13.3	10.4	8.6	6.7	5.9	5.5	6.4	8.4	10.8	12.3	
	5% WB	MCDBR	17.7	18.1	16.7	14.8	14.3	12.4	13.0	12.3	12.4	13.3	14.3	16.4	
		MCWBR	13.3	14.7	13.9	10.0	8.6	6.5	5.7	5.4	6.4	8.7	11.3	12.6	
Clear-Sky Solar Irradiance	taub	0.377	0.426	0.490	0.540	0.564	0.596	0.587	0.578	0.515	0.484	0.423	0.379		
	taud	2.254	2.115	1.959	1.880	1.880	1.870	1.939	1.953	2.094	2.102	2.194	2.274		
	Ebn at Noon	254	253	246	241	237	229	230	229	237	231	235	245		
	Edh at Noon	34	43	55	62	63	64	59	57	48	43	36	31		
All-Sky Solar Radiation	RadAvg	900	1059	1313	1528	1637	1429	1567	1575	1224	973	810	785		
	RadStd	64	106	83	154	174	137	265	145	112	107	78	62		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (39 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page